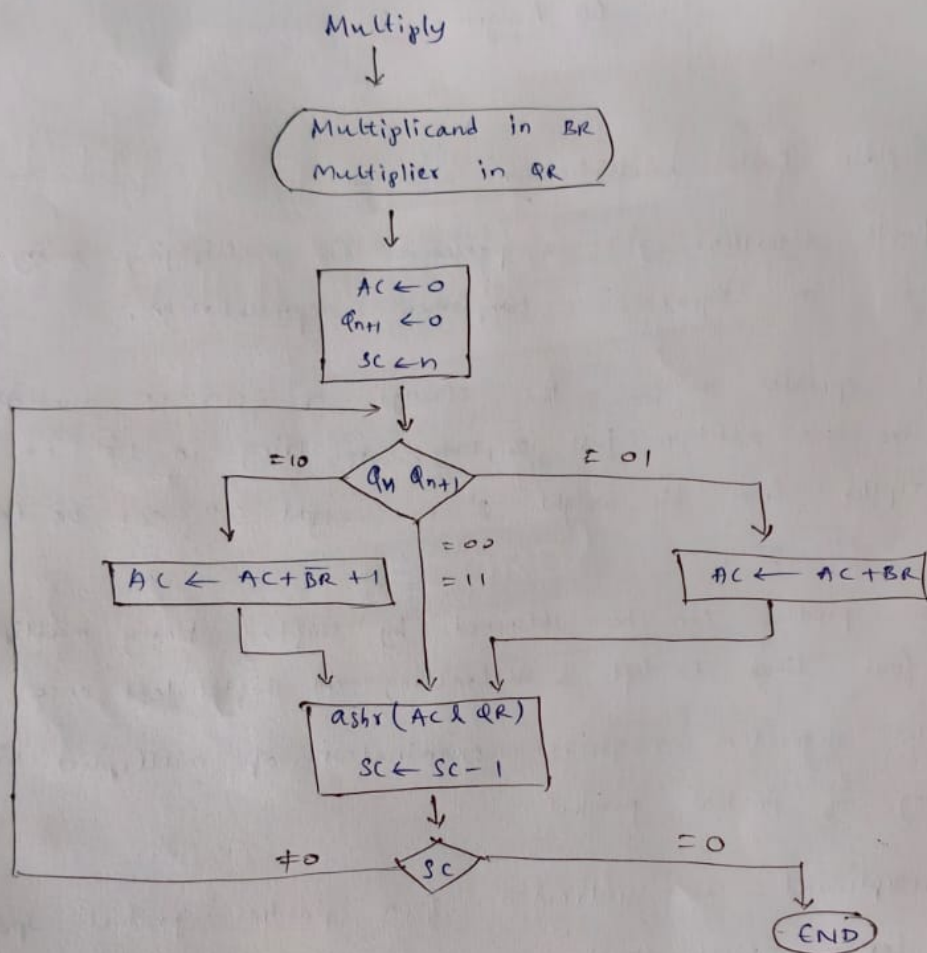


CO - Assignment - II

1Q. Explain Booth multiplication algorithm.

- Booth algorithm gives a procedure for multiplying binary integers in signed-2's complement representation.
- It operates on fact that strings of 2's in multiplier require no addition, just shifting, & string of 1's in multiplier from bit weight 2^k to weight 2^m can be treated as $2^{k+1} - 2^m$.
- Thus product can be obtained by shifting binary multiplicand m four times to left & subtracting m shifted left once.
- Booth algorithm requires examination of multiplier bits & shifting of partial product.
- Multiplicand is subtracted from partial product upon encountering first least significant in a string 1's in multiplier.
- Multiplicand is added to partial product upon encountering the first 0.
- The partial product doesn't change when multiplier bit is identical to previous bit.
- Algorithm works for (+ve) or (-ve) multipliers in 2's representation.



Table

Q_n	Q_{n+1}	BR = 10111 $\overline{BR} + 1 = 01001$	AC	QR	Q_{n+1}	SC
		Initial Subtract BR	00000 01001 01001	0 1011	0	101
1	0	ashr ashr add BR	00100 00010 10111 11001	11001 01100	1 1	100 011
0	0	ashr ashr Sub BR	111000 11110 01001 00111			
1	0	ashr	00011	10101	1	000

1 Q. Explain Microprogram Sequencer?

Microprogram sequencer is part of CPU (control unit) that generates address of next microinstruction to be executed, helps in sequencing control signals for execution of machine instructions.

Functions :-

- * Determines address of next microinstruction.
- * Supports branching, & returns
- * Controls flow of microinstructions.

Components :-

Control Address register :- Holds address of current microinstruction.

Control memory :- Stores microinstruction

Control Buffer Register :- Holds current microinstruction.

Sequencer logic :- Determines next CAR value:

- Increment
- Branch address
- Return address
- External input.

Types of sequencing :-

- Sequential : Next is address $CAR+1$
- Branching : Based on conditions
- Mapping : Jump to routine based on opcode.

⇒ 34. Binary adder in micro operations.

• Basic arithmetic microoperations are:

* addition

* subtraction

* increment

* decrement

* shift.

• Addition microoperation

$$R_3 \leftarrow R_1 + R_2$$

• Subtraction microoperation

$$R_3 \leftarrow R_1 - R_2$$

• One's complement:

$$R_2 \leftarrow \bar{R}_2$$

• Decrement

microoperation:

$$R_2 \leftarrow R_2 - 1$$

• Two's complement:

$$R_2 \leftarrow \bar{R}_2 + 1$$

• Increment microoperation:

$$R_2 \leftarrow R_2 + 1$$

⇒ Summary:-

operations

Description

$$R_3 \leftarrow R_1 + R_2$$

Contents of R_1 plus R_2 to R_3

$$R_3 \leftarrow R_1 - R_2$$

Contents of R_1 minus R_2 to R_3

$$R_2 \leftarrow \bar{R}_2$$

Complement of R_2

$$R_2 \leftarrow \bar{R}_2 + 1$$

2's complement Contents of R_2 (neg)

$$R_3 \leftarrow R_1 + R_2 + 1$$

Subtraction

$$R_1 \leftarrow R_1 + 1$$

Increment

$$R_1 \leftarrow R_1 - 1$$

Decrement

⇒ Binary adder:-

